covid | the 10 largest locations without Florida pooled | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

Fixed MultiQT
Leaky MultiQT
Delphi REF-op (covid prod)
Delphi sqrt 0.018 (flu prod)
Fixed MultiQT, adaptive rate
Leaky MultiQT, adaptive rate
Sqrt tracker, adaptive rate

Log1p tracker

Fixed MultiQT (log1p tracker)
Leaky MultiQT (log1p tracker)
Fixed MultiQT (log1p tracker), adaptive rate
Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

Rate grid: equal
Rate grid: learned, no sharing
Rate grid: learned, common weights
Rate grid: learned + sharing
Rate grid: learned + sharing, common weights
Rate grid: sharing + outer
Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

First start only
Weekly starts: equal
Weekly starts: learned + sharing
Weekly starts: most recent, learned + sharing
Weekly starts: most recent, slow + fast, learned + sharing
Starts equal + outer
Starts learned + sharing + outer

Score-based experts

Score experts: equal weights
Score experts: learned weights

Studentized median

Studentized median (residual RMS)
Studentized median (residual RMS), median start
Studentized median (base width)
Median, constant sigma
Median, constant sigma, median start

Studentized full base

Studentized full base (base width)
Studentized full base (level RMS)
Studentized full base (PIT)

Power scale

Studentized full base (power scale)
Power scale: rate grid, equal
Power scale: rate grid, learned, common weights
Power scale: most recent starts, equal
Power scale: most recent starts, learned + sharing

t-PIT scale

Studentized full base (t-PIT scale)
t-PIT scale: rate grid, equal
t-PIT scale: rate grid, learned, common weights
t-PIT scale: most recent starts, equal
t-PIT scale: most recent starts, learned + sharing

Spread scale

Studentized full base (spread scale)
Spread scale: rate grid, equal
Spread scale: rate grid, learned, common weights
Spread scale: most recent starts, equal
Spread scale: most recent starts, learned + sharing

-60% -40% -20% 0% · 0 0.1 0.2

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
- Delphi REF-op (covid prod)
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- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: learned, no sharing
- Rate grid: learned, common weights
- Rate grid: learned + sharing
- Rate grid: learned + sharing, common weights
- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: learned + sharing
- Weekly starts: most recent, learned + sharing
- Weekly starts: most recent, slow + fast, learned + sharing
- Starts equal + outer
- Starts learned + sharing + outer

Score-based experts

- Score experts: equal weights
- Score experts: learned weights

Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss bars: -50% 0% 50% · Coverage error: 0 0.1 0.2 0.3

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

covid | California (06) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 | Feedback delay 1 | Feedback re-run

Loss reduction: overall | Loss reduction: up to peak (weeks -7 to 0) | Loss reduction: after peak (weeks 1 to 8) | Coverage error: overall | Coverage error: up to peak (weeks -7 to 0) | Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
- Delphi REF-op (covid prod)
- Delphi sqrt 0.018 (flu prod)
- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: learned, no sharing
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- Rate grid: learned + sharing
- Rate grid: learned + sharing, common weights
- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: learned + sharing
- Weekly starts: most recent, learned + sharing
- Weekly starts: most recent, slow + fast, learned + sharing
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- Starts learned + sharing + outer

Score-based experts

- Score experts: equal weights
- Score experts: learned weights

Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss bars: -100% -50% 0% 50% 100%. Coverage error: 0 0.1 0.2 0.3 0.4.

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
- Delphi REF-op (covid prod)
- Delphi sqrt 0.018 (flu prod)
- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: learned, no sharing
- Rate grid: learned, common weights
- Rate grid: learned + sharing
- Rate grid: learned + sharing, common weights
- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: learned + sharing
- Weekly starts: most recent, learned + sharing
- Weekly starts: most recent, slow + fast, learned + sharing
- Starts equal + outer
- Starts learned + sharing + outer

Score-based experts

- Score experts: equal weights
- Score experts: learned weights

Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss panels: -50% 0% 50% · Coverage panels: 0 0.1 0.2

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 | Feedback delay 1 | Feedback re-run

Loss reduction: overall | Loss reduction: up to peak (weeks -7 to 0) | Loss reduction: after peak (weeks 1 to 8) | Coverage error: overall | Coverage error: up to peak (weeks -7 to 0) | Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
- Delphi REF-op (covid prod)
- Delphi sqrt 0.018 (flu prod)
- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: leamed, no sharing
- Rate grid: leamed, common weights
- Rate grid: leamed + sharing
- Rate grid: leamed + sharing, common weights
- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: leamed + sharing
- Weekly starts: most recent, leamed + sharing
- Weekly starts: most recent, slow + fast, leamed + sharing
- Starts equal + outer
- Starts leamed + sharing + outer

Score-based experts

- Score experts: equal weights
- Score experts: learned weights

Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, leamed, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, leamed + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, leamed, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, leamed + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, leamed, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, leamed + sharing

Loss panels: -80% -60% -40% -20% 0% 20%. Coverage error panels: 0 0.05 0.1 0.15 0.2.

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 | Feedback delay 1 | Feedback re-run

Loss reduction: overall | Loss reduction: up to peak (weeks -7 to 0) | Loss reduction: after peak (weeks 1 to 8) | Coverage error: overall | Coverage error: up to peak (weeks -7 to 0) | Coverage error: after peak (weeks 1 to 8)

Single trackers

Fixed MultiQT

Leaky MultiQT

Delphi REF-op (covid prod)

Delphi sqrt 0.018 (flu prod)

Fixed MultiQT, adaptive rate

Leaky MultiQT, adaptive rate

Sqrt tracker, adaptive rate

Log1p tracker

Fixed MultiQT (log1p tracker)

Leaky MultiQT (log1p tracker)

Fixed MultiQT (log1p tracker), adaptive rate

Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

Rate grid: equal

Rate grid: learned, no sharing

Rate grid: learned, common weights

Rate grid: learned + sharing

Rate grid: learned + sharing, common weights

Rate grid: sharing + outer

Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

First start only

Weekly starts: equal

Weekly starts: learned + sharing

Weekly starts: most recent, learned + sharing

Weekly starts: most recent, slow + fast, learned + sharing

Starts equal + outer

Starts learned + sharing + outer

Score-based experts

Score experts: equal weights

Score experts: learned weights

Studentized median

Studentized median (residual RMS)

Studentized median (residual RMS), median start

Studentized median (base width)

Median, constant sigma

Median, constant sigma, median start

Studentized full base

Studentized full base (base width)

Studentized full base (level RMS)

Studentized full base (PIT)

Power scale

Studentized full base (power scale)

Power scale: rate grid, equal

Power scale: rate grid, learned, common weights

Power scale: most recent starts, equal

Power scale: most recent starts, learned + sharing

t-PIT scale

Studentized full base (t-PIT scale)

t-PIT scale: rate grid, equal

t-PIT scale: rate grid, learned, common weights

t-PIT scale: most recent starts, equal

t-PIT scale: most recent starts, learned + sharing

Spread scale

Studentized full base (spread scale)

Spread scale: rate grid, equal

Spread scale: rate grid, learned, common weights

Spread scale: most recent starts, equal

Spread scale: most recent starts, learned + sharing

-80% -60% -40% -20% 0% 20% | -80% -60% -40% -20% 0% 20% | -80% -60% -40% -20% 0% 20% | 0 0.05 0.1 0.15 0.2 0.25 | 0 0.05 0.1 0.15 0.2 0.25 | 0 0.05 0.1 0.15 0.2 0.25

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | the 10 largest locations without Florida pooled | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

Fixed MultiQT
Leaky MultiQT
Delphi REF-op (covid prod)
Delphi sqrt 0.018 (flu prod)
Fixed MultiQT, adaptive rate
Leaky MultiQT, adaptive rate
Sqrt tracker, adaptive rate

Log1p tracker

Fixed MultiQT (log1p tracker)
Leaky MultiQT (log1p tracker)
Fixed MultiQT (log1p tracker), adaptive rate
Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

Rate grid: equal
Rate grid: learned, no sharing
Rate grid: learned, common weights
Rate grid: learned + sharing
Rate grid: learned + sharing, common weights
Rate grid: sharing + outer
Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

First start only
Weekly starts: equal
Weekly starts: learned + sharing
Weekly starts: most recent, learned + sharing
Weekly starts: most recent, slow + fast, learned + sharing
Starts equal + outer
Starts learned + sharing + outer

Score-based experts

Score experts: equal weights
Score experts: learned weights

Studentized median

Studentized median (residual RMS)
Studentized median (residual RMS), median start
Studentized median (base width)
Median, constant sigma
Median, constant sigma, median start

Studentized full base

Studentized full base (base width)
Studentized full base (level RMS)
Studentized full base (PIT)

Power scale

Studentized full base (power scale)
Power scale: rate grid, equal
Power scale: rate grid, learned, common weights
Power scale: most recent starts, equal
Power scale: most recent starts, learned + sharing

t-PIT scale

Studentized full base (t-PIT scale)
t-PIT scale: rate grid, equal
t-PIT scale: rate grid, learned, common weights
t-PIT scale: most recent starts, equal
t-PIT scale: most recent starts, learned + sharing

Spread scale

Studentized full base (spread scale)
Spread scale: rate grid, equal
Spread scale: rate grid, learned, common weights
Spread scale: most recent starts, equal
Spread scale: most recent starts, learned + sharing

Loss reduction: -60% -40% -20% 0%
Coverage error: 0 0.1 0.2 0.3

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 | Feedback delay 1 | Feedback re-run

Loss reduction: overall | Loss reduction: up to peak (weeks -7 to 0) | Loss reduction: after peak (weeks 1 to 8) | Coverage error: overall | Coverage error: up to peak (weeks -7 to 0) | Coverage error: after peak (weeks 1 to 8)

Single trackers

Fixed MultiQT
Leaky MultiQT
Delphi REF-op (covid prod)
Delphi sqrt 0.018 (flu prod)
Fixed MultiQT, adaptive rate
Leaky MultiQT, adaptive rate
Sqrt tracker, adaptive rate

Log1p tracker

Fixed MultiQT (log1p tracker)
Leaky MultiQT (log1p tracker)
Fixed MultiQT (log1p tracker), adaptive rate
Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

Rate grid: equal
Rate grid: leamed, no sharing
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Rate grid: leamed + sharing
Rate grid: leamed + sharing, common weights
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Weekly starts: immediate admission

First start only
Weekly starts: equal
Weekly starts: leamed + sharing
Weekly starts: most recent, leamed + sharing
Weekly starts: most recent, slow + fast, leamed + sharing
Starts equal + outer
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Score-based experts

Score experts: equal weights
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Studentized median

Studentized median (residual RMS)
Studentized median (residual RMS), median start
Studentized median (base width)
Median, constant sigma
Median, constant sigma, median start

Studentized full base

Studentized full base (base width)
Studentized full base (level RMS)
Studentized full base (PIT)

Power scale

Studentized full base (power scale)
Power scale: rate grid, equal
Power scale: rate grid, leamed, common weights
Power scale: most recent starts, equal
Power scale: most recent starts, leamed + sharing

t-PIT scale

Studentized full base (t-PIT scale)
t-PIT scale: rate grid, equal
t-PIT scale: rate grid, leamed, common weights
t-PIT scale: most recent starts, equal
t-PIT scale: most recent starts, leamed + sharing

Spread scale

Studentized full base (spread scale)
Spread scale: rate grid, equal
Spread scale: rate grid, leamed, common weights
Spread scale: most recent starts, equal
Spread scale: most recent starts, leamed + sharing

-50% 0% 50% | -50% 0% 50% | -50% 0% 50% | 0 0.1 0.2 0.3 0.4 | 0 0.1 0.2 0.3 0.4 | 0 0.1 0.2 0.3 0.4

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
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Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | California (06) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
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- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: learned, no sharing
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- Rate grid: learned + sharing
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- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: learned + sharing
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- Weekly starts: most recent, slow + fast, learned + sharing
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Score-based experts

- Score experts: equal weights
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Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss bars: -50% 0% 50% · Coverage error: 0 0.05 0.1 0.15 0.2 0.25

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
- Leaky MultiQT
- Delphi REF-op (covid prod)
- Delphi sqrt 0.018 (flu prod)
- Fixed MultiQT, adaptive rate
- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

Rate-grid experts

- Rate grid: equal
- Rate grid: learned, no sharing
- Rate grid: learned, common weights
- Rate grid: learned + sharing
- Rate grid: learned + sharing, common weights
- Rate grid: sharing + outer
- Rate grid: sharing, common weights + outer

Weekly starts: immediate admission

- First start only
- Weekly starts: equal
- Weekly starts: learned + sharing
- Weekly starts: most recent, learned + sharing
- Weekly starts: most recent, slow + fast, learned + sharing
- Starts equal + outer
- Starts learned + sharing + outer

Score-based experts

- Score experts: equal weights
- Score experts: learned weights

Studentized median

- Studentized median (residual RMS)
- Studentized median (residual RMS), median start
- Studentized median (base width)
- Median, constant sigma
- Median, constant sigma, median start

Studentized full base

- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

Power scale

- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss bars: -20% -10% 0% 10% 20% · Coverage error: 0 0.05 0.1 0.15 0.2 0.25

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 · Feedback delay 1 · Feedback re-run

Loss reduction: overall · Loss reduction: up to peak (weeks -7 to 0) · Loss reduction: after peak (weeks 1 to 8) · Coverage error: overall · Coverage error: up to peak (weeks -7 to 0) · Coverage error: after peak (weeks 1 to 8)

Single trackers

- Fixed MultiQT
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- Leaky MultiQT, adaptive rate
- Sqrt tracker, adaptive rate

Log1p tracker

- Fixed MultiQT (log1p tracker)
- Leaky MultiQT (log1p tracker)
- Fixed MultiQT (log1p tracker), adaptive rate
- Leaky MultiQT (log1p tracker), adaptive rate

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- Studentized full base (base width)
- Studentized full base (level RMS)
- Studentized full base (PIT)

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- Studentized full base (power scale)
- Power scale: rate grid, equal
- Power scale: rate grid, learned, common weights
- Power scale: most recent starts, equal
- Power scale: most recent starts, learned + sharing

t-PIT scale

- Studentized full base (t-PIT scale)
- t-PIT scale: rate grid, equal
- t-PIT scale: rate grid, learned, common weights
- t-PIT scale: most recent starts, equal
- t-PIT scale: most recent starts, learned + sharing

Spread scale

- Studentized full base (spread scale)
- Spread scale: rate grid, equal
- Spread scale: rate grid, learned, common weights
- Spread scale: most recent starts, equal
- Spread scale: most recent starts, learned + sharing

Loss reduction: -50% 0% 50% · Coverage error: 0 0.1 0.2 0.3 0.4

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.

flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | loss in counts per 100k residents | every method under each feedback design

Feedback delay 0 | Feedback delay 1 | Feedback re-run

Loss reduction: overall | Loss reduction: up to peak (weeks -7 to 0) | Loss reduction: after peak (weeks 1 to 8) | Coverage error: overall | Coverage error: up to peak (weeks -7 to 0) | Coverage error: after peak (weeks 1 to 8)

Single trackers

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Log1p tracker

Fixed MultiQT (log1p tracker)
Leaky MultiQT (log1p tracker)
Fixed MultiQT (log1p tracker), adaptive rate
Leaky MultiQT (log1p tracker), adaptive rate

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Weekly starts: equal
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Studentized median (base width)
Median, constant sigma
Median, constant sigma, median start

Studentized full base

Studentized full base (base width)
Studentized full base (level RMS)
Studentized full base (PIT)

Power scale

Studentized full base (power scale)
Power scale: rate grid, equal
Power scale: rate grid, learned, common weights
Power scale: most recent starts, equal
Power scale: most recent starts, learned + sharing

t-PIT scale

Studentized full base (t-PIT scale)
t-PIT scale: rate grid, equal
t-PIT scale: rate grid, learned, common weights
t-PIT scale: most recent starts, equal
t-PIT scale: most recent starts, learned + sharing

Spread scale

Studentized full base (spread scale)
Spread scale: rate grid, equal
Spread scale: rate grid, learned, common weights
Spread scale: most recent starts, equal
Spread scale: most recent starts, learned + sharing

Loss bars: -50% 0% 50%
Coverage error: 0 0.1 0.2 0.3

Values and axes as in the evaluation tree. Loss bars: 100 * (1 - pooled loss / pooled loss of the base) over the page's cases of a design, each case's mean loss weighted by its weeks; positive = smaller loss.
Numbers right of the loss panels: share of the cases in which the method's loss is below the base's (at least 5 cases). Coverage error: mean over the levels of |coverage - alpha|, weighted by the weeks; dashed: the base.
Top to bottom within a method: feedback delay 0, delay 1, re-run (darker to lighter). Each design uses the settings tuned with it.